

8.10 Spectral Mapping, Products in Both Orders, Left Eigenvectors

Once the eigenvalues of T are known, what can we say about the eigenvalues of T^2 , of $T^2 - 3T + I$, or of any polynomial in T ? And when two matrices are multiplied in the two possible orders, how are the eigenvalues of AB and BA related? This section answers both questions using the triangular forms of Section 7 and the block matrices of Chapter 7. Along the way it settles two debts: Chapter 6 showed $p_{AB} = p_{BA}$ only when A is invertible, and Chapter 7 proved that vanishing traces of all powers force a matrix to be nilpotent by a long computation, with no explanation of why. We end with left eigenvectors, the eigenvectors of the transpose, which pair with ordinary eigenvectors in a useful way and prepare the Markov chains of Section 11.

8.10.1 Eigenvalues of a polynomial in T

One half of the answer needs no theory. If $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $T^k\mathbf{v} = \lambda^k\mathbf{v}$ for every k , and so $q(T)\mathbf{v} = q(\lambda)\mathbf{v}$ for every $q \in F[x]$ (Exercise (B3: Eigenvalues of powers and inverses)). So q **maps eigenvalues of T to eigenvalues of $q(T)$** , over every field and in every dimension.

The other half, that **every** eigenvalue of $q(T)$ arises this way, can fail. Part (c) of Exercise (B3: Eigenvalues of powers and inverses) already met the rotation $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ on $\mathbb{R}^2$: it has no real eigenvalue, while $R^2 = -I$ has the eigenvalue -1 . The eigenvalues of R that would map to -1 , namely $\pm i$, are missing from $\mathbb{R}$. So we expect the full statement to need a characteristic polynomial that splits. With that hypothesis we even get the multiplicities, because a triangular form shows all the eigenvalues at once on its diagonal.

The eigenvalues of $q(T)$ are the values of q at the eigenvalues of T , counted with multiplicity.

Theorem 8.10.1 — Spectral Mapping Theorem

Let V be a finite-dimensional vector space over F with $\dim V = n \geq 1$, and let $T \in \mathcal{L}(V)$. Suppose p_T splits over F , say

$$p_T(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_n),$$

where each eigenvalue appears as often as its algebraic multiplicity. Then for every $q \in F[x]$,

$$p_{q(T)}(x) = (x - q(\lambda_1))(x - q(\lambda_2)) \cdots (x - q(\lambda_n)),$$

and consequently

$$\text{spec}(q(T)) = q(\text{spec}(T)) = \{q(\lambda) : \lambda \in \text{spec}(T)\}.$$

In particular this holds for every operator on a non-zero finite-dimensional complex space, and for every $A \in M_n(\mathbb{C})$.

Idea. Choose a basis in which T is upper triangular (Section 7). Its diagonal is $\lambda_1, \dots, \lambda_n$. Products and sums of upper triangular matrices are upper triangular, and their diagonals are computed position by position, so $q([T])$ is upper triangular with diagonal

$q(\lambda_1), \dots, q(\lambda_n)$. The characteristic polynomial of a triangular matrix is read off its diagonal.

Proof. By Theorem 8.7.2 there is a basis $\mathcal{B}$ of V such that $U = [T]_{\mathcal{B}}$ is upper triangular. By Theorem 8.7.4, its diagonal entries are $\lambda_1, \dots, \lambda_n$ in some order; renumbering the λ_i , we may assume $u_{ii} = \lambda_i$.

Claim. If $X, Y \in M_n(F)$ are upper triangular, then so are $X + Y$, cX for $c \in F$, and XY , with diagonal entries $x_{ii} + y_{ii}$, cx_{ii} and $x_{ii}y_{ii}$.

Proof. The statements for $X + Y$ and cX are entrywise. For the product, $(XY)_{ij} = \sum_{k=1}^n x_{ik}y_{kj}$ (Definition 0.9.9). If $i > j$, every k has $k < i$ or $k \geq i > j$, so $x_{ik} = 0$ or $y_{kj} = 0$, and the sum is 0. If $i = j$, the only term that can be non-zero has $k \geq i$ and $k \leq i$, so $(XY)_{ii} = x_{ii}y_{ii}$. ■

By the claim and induction on k , U^k is upper triangular with diagonal entries $\lambda_1^k, \dots, \lambda_n^k$ (for $k = 0$, $U^0 = I$). Writing $q = \sum_k a_k x^k$, the claim again shows that $q(U) = \sum_k a_k U^k$ is upper triangular with diagonal entries $\sum_k a_k \lambda_i^k = q(\lambda_i)$. By Corollary 3.6.3, $[q(T)]_{\mathcal{B}} = q(U)$, so by Definition 6.10.2 and the examples after Definition 6.10.1,

$$p_{q(T)} = p_{q(U)} = \prod_{i=1}^n (x - q(\lambda_i)).$$

For the spectra: by Theorem 8.2.1, $\mu \in F$ is an eigenvalue of $q(T)$ if and only if $p_{q(T)}(\mu) = \prod_i (\mu - q(\lambda_i)) = 0$, that is, if and only if $\mu = q(\lambda_i)$ for some i , since a product of non-zero field elements is non-zero. The λ_i are exactly the eigenvalues of T , each listed at least once (Theorem 8.2.1). Hence $\text{spec}(q(T)) = \{q(\lambda) : \lambda \in \text{spec}(T)\}$. Over $\mathbb{C}$, p_T always splits by Corollary 5.5.5. ■

The formula for $p_{q(T)}$ carries more than the spectrum. The algebraic multiplicity of μ as an eigenvalue of $q(T)$ is the number of indices i with $q(\lambda_i) = \mu$ (Lemma 5.4.1), so several eigenvalues of T can merge into one eigenvalue of $q(T)$, and their multiplicities add. Combined with Theorem 8.3.3, applied to $q(T)$, whose characteristic polynomial splits by the theorem, we get the most used consequence.

Corollary 8.10.2 — Trace and Determinant of a Polynomial in T

Under the hypotheses of Theorem 8.10.1, for every $q \in F[x]$,

$$\text{tr } q(T) = q(\lambda_1) + \dots + q(\lambda_n), \quad \det q(T) = q(\lambda_1) \cdots q(\lambda_n).$$

In particular $\text{tr } T^k = \lambda_1^k + \dots + \lambda_n^k$ for every $k \in \mathbb{N}$.

Proof. By Theorem 8.10.1, $p_{q(T)} = \prod_i (x - q(\lambda_i))$ splits. Group equal values: if $\mu_1, \dots, \mu_s$ are the distinct values among the $q(\lambda_i)$, then μ_j occurs exactly $a_{q(T)}(\mu_j)$ times in the list

(Lemma 5.4.1). By Theorem 8.3.3 applied to $q(T)$, $\text{tr } q(T) = \sum_j a_{q(T)}(\mu_j) \mu_j = \sum_i q(\lambda_i)$ and $\det q(T) = \prod_j \mu_j^{a_{q(T)}(\mu_j)} = \prod_i q(\lambda_i)$. Take $q = x^k$ for the last statement. ■

Example. (A Polynomial in a Non-Diagonalizable Matrix)

Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ -3 & 3 & 2 \end{pmatrix} \in M_3(\mathbb{R})$, for which Exercise (B1: Diagonalize or show impossible) found $p_A = (x+1)(x-2)^2$ and showed that A is not diagonalizable. Without computing any eigenvectors, find p_{A^2} , $\text{tr } A^2$ and $\det A^2$. Then find $p_{q(A)}$ for $q = x^2 - x - 2$, and decide whether $q(A) = 0$.

Solution. The square. p_A splits over $\mathbb{R}$ with the list $\lambda_1 = -1$, $\lambda_2 = \lambda_3 = 2$. By Theorem 8.10.1 with $q = x^2$, the list for A^2 is 1, 4, 4, so

$$p_{A^2} = (x-1)(x-4)^2, \quad \text{tr } A^2 = 1 + 4 + 4 = 9, \quad \det A^2 = 1 \cdot 4 \cdot 4 = 16$$

by Corollary 8.10.2. *Check.* Multiplying out,

$$A^2 = \begin{pmatrix} 0 & 4 & 4 \\ -1 & 5 & 4 \\ -3 & 3 & 4 \end{pmatrix},$$

whose trace is $0 + 5 + 4 = 9$; and $\det A = (-1) \cdot 2 \cdot 2 = -4$ by Theorem 8.3.3, so $\det A^2 = (\det A)^2 = 16$ by Theorem 6.5.4.

The polynomial q . Here $q = (x-2)(x+1)$, so $q(-1) = 0$ and $q(2) = 0$. The list for $q(A)$ is 0, 0, 0, and $p_{q(A)} = x^3$: the only eigenvalue of $q(A)$ is 0. Yet $q(A) \neq 0$. Indeed

$$q(A) = A^2 - A - 2I = \begin{pmatrix} -3 & 3 & 3 \\ -3 & 3 & 3 \\ 0 & 0 & 0 \end{pmatrix},$$

and one checks $q(A)^2 = 0$. The spectral mapping theorem sees only eigenvalues, and a non-zero matrix can have 0 as its only eigenvalue. That $q(A) \neq 0$ here is no accident: if $q(A) = 0$, the minimal polynomial of A would divide (Theorem 8.5.1) $(x-2)(x+1)$, which has distinct roots, and A would be diagonalizable by Theorem 8.8.2.

Warning. Without splitting, spectral mapping fails. For the real rotation R above, $\text{spec}(R) = \emptyset$ in $\mathbb{R}$, so $q(\text{spec}(R)) = \emptyset$ for $q = x^2$, while $\text{spec}(R^2) = \text{spec}(-I) = \{-1\}$. Over $\mathbb{C}$ the theorem is restored: $\text{spec}(R) = \{i, -i\}$, both square to -1 , and $p_{R^2} = (x+1)^2$. Also, the theorem is about eigenvalues and algebraic multiplicities only: $T = \text{diag}(1, -1)$ has two one-dimensional eigenspaces, while $T^2 = I$ has the single eigenspace $\mathbb{R}^2$.

Quick check. Let $A \in M_2(\mathbb{C})$ have eigenvalues i and $-i$. Find p_{A^2} and $\det(A^2 + I)$.

Solution. The list for A is $i, -i$ (two distinct roots of the degree-2 polynomial p_A). By

Theorem 8.10.1 with $q = x^2$, the list for A^2 is $-1, -1$, so $p_{A^2} = (x+1)^2$. With $q = x^2 + 1$, **Corollary 8.10.2** gives $\det(A^2 + I) = q(i)q(-i) = 0 \cdot 0 = 0$.

8.10.2 Traces of powers detect nilpotency

Chapter 7 proved, in Exercise (C2: When A commutes with $[A, B]$), that a matrix C over a field of characteristic 0 with $\operatorname{tr} C^k = 0$ for every $k \geq 1$ is nilpotent, by a careful argument with annihilating polynomials that gave no reason for the result. Here is the view through eigenvalues, for complex matrices. The traces of the powers are the **power sums** $\lambda_1^k + \cdots + \lambda_n^k$ of the eigenvalues (**Corollary 8.10.2**), and a nilpotent matrix is one whose eigenvalues are all 0. So the question becomes: if all power sums of n complex numbers vanish, must the numbers vanish? A Vandermonde determinant says yes.

Theorem 8.10.3 — Nilpotent Matrices via Eigenvalues and Traces

Let $n \geq 1$ and $A \in M_n(\mathbb{C})$. The following are equivalent:

- (a) A is nilpotent, that is, $A^s = 0$ for some $s \geq 1$;
- (b) $\operatorname{spec}(A) = \{0\}$;
- (c) $p_A = x^n$;
- (d) $A^n = 0$;
- (e) $\operatorname{tr} A^k = 0$ for $k = 1, 2, \dots, n$.

The same holds for $A \in M_n(F)$ when F is a subfield of $\mathbb{C}$, such as $\mathbb{Q}$ or $\mathbb{R}$, with $\operatorname{spec}(A)$ in (b) taken over $\mathbb{C}$.

Idea. Plan: (a) $\Rightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (d) $\Rightarrow$ (a), and (c) $\Rightarrow$ (e) $\Rightarrow$ (b). All but one arrow are short. For (e) $\Rightarrow$ (b), suppose some eigenvalue is non-zero. Group the non-zero eigenvalues into distinct values $\mu_1, \dots, \mu_r$ with multiplicities $m_1, \dots, m_r \geq 1$. The vanishing traces say $\sum_j m_j \mu_j^k = 0$ for $k = 1, \dots, r$: a square linear system for the vector $(m_1, \dots, m_r)$, whose matrix is a Vandermonde matrix times the invertible $\operatorname{diag}(\mu_j)$. So the m_j would be 0, but they are positive integers, and positive integers are non-zero in $\mathbb{C}$.

Proof.

- (a) $\Rightarrow$ (b). This is Exercise (C2: Nilpotent operators) (a), applied to T_A on $\mathbb{C}^n \neq \{0\}$.
- (b) $\Rightarrow$ (c). By Corollary 5.5.5, $p_A = (x - z_1) \cdots (x - z_n)$ with $z_i \in \mathbb{C}$. Each z_i is an eigenvalue (Theorem 8.2.1), hence 0, so $p_A = x^n$.
- (c) $\Rightarrow$ (d). By Corollary 8.7.3, $A = PUP^{-1}$ with P invertible and U upper triangular, and by Theorem 8.7.4 the diagonal entries of U are the roots of $p_A = x^n$, all 0. Then $U\mathbf{e}_1 = \mathbf{0}$ and $U\mathbf{e}_j \in \operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_{j-1})$ for $j \geq 2$, since column j of U has zeros from row j down. By induction on j , $U^j \mathbf{e}_j = \mathbf{0}$, hence $U^n \mathbf{e}_j = U^{n-j} U^j \mathbf{e}_j = \mathbf{0}$ for every j , and $U^n = 0$. By Proposition 3.7.6 (c), $A^n = P U^n P^{-1} = 0$.
- (d) $\Rightarrow$ (a). Take $s = n$.
- (e) $\Rightarrow$ (e). The list of eigenvalues of A with multiplicity is $0, \dots, 0$, so by **Corollary 8.10.2**, $\operatorname{tr} A^k = 0^k + \cdots + 0^k = 0$ for every $k \geq 1$.

(f) $\Rightarrow$ (b). By Corollary 5.5.5, p_A splits; let $\lambda_1, \dots, \lambda_n$ be the list of its roots with multiplicity. Since $n \geq 1$, $\text{spec}(A)$ is non-empty, so it suffices to show that every $\lambda_i = 0$. Suppose not, let $\mu_1, \dots, \mu_r$ ($1 \leq r \leq n$) be the distinct non-zero values among the λ_i , and let $m_j \geq 1$ be the number of indices i with $\lambda_i = \mu_j$. For $1 \leq k \leq r$, the zero eigenvalues contribute $0^k = 0$, so by Corollary 8.10.2 and (e),

$$\sum_{j=1}^r m_j \mu_j^k = \text{tr } A^k = 0.$$

In matrix form, $W\mathbf{m} = \mathbf{0}$, where $\mathbf{m} = (m_1, \dots, m_r) \in \mathbb{C}^r$ and $W \in M_r(\mathbb{C})$ has (k, j) -entry $\mu_j^k = \mu_j^{k-1} \mu_j$. Hence $W = V(\mu_1, \dots, \mu_r)^\top \text{diag}(\mu_1, \dots, \mu_r)$, where $V(\mu_1, \dots, \mu_r)$ is the Vandermonde matrix with (j, k) -entry μ_j^{k-1} (Theorem 6.8.1). The μ_j are distinct, so V is invertible (Corollary 6.8.2), and so is $V^\top$, with inverse $(V^{-1})^\top$ (Theorem 0.9.5). The μ_j are non-zero, so $\text{diag}(\mu_1, \dots, \mu_r)$ is invertible. Therefore W is invertible, and $\mathbf{m} = W^{-1}\mathbf{0} = \mathbf{0}$. But m_1 is a positive integer, and $m_1 \neq 0$ in $\mathbb{C}$. This contradiction shows that every $\lambda_i = 0$, so $\text{spec}(A) = \{0\}$.

For a subfield $F \subseteq \mathbb{C}$, regard A as a complex matrix. Its powers, their traces, its characteristic polynomial and the condition $A^s = 0$ are computed by the same arithmetic in F and in $\mathbb{C}$, so (a), (c), (d) and (e) mean the same thing over F and over $\mathbb{C}$, and the equivalences carry over. ■

In the setting of Exercise (C2: When A commutes with $[A, B]$), where $C = [A, B]$ commutes with A , part (a) of that exercise gives $\text{tr } C^k = 0$ for all $k \geq 1$, and for complex matrices the theorem now shows at once that $C^n = 0$. That exercise remains the route for a general field of characteristic 0.

Warning. The characteristic matters. Over $\mathbb{F}_p$, the identity $I_p \in M_p(\mathbb{F}_p)$ has $\text{tr } I_p^k = p \cdot 1 = 0$ for every k , yet $I_p^k = I_p \neq 0$. The proof above breaks exactly where it used that the positive integer m_1 is non-zero in the field: here the multiplicity $m_1 = p$ of the eigenvalue 1 is 0 in $\mathbb{F}_p$.

8.10.3 AB and BA

Chapter 6, in Exercise (B3: AB and BA), showed that $p_{AB} = p_{BA}$ for square A and B **when A is invertible**, because then $BA = A^{-1}(AB)A$ is similar to AB . Without invertibility the two products need not be similar, and for rectangular A and B they do not even have the same size: if A is $m \times n$ and B is $n \times m$, then AB is $m \times m$ and BA is $n \times n$. Their characteristic polynomials have degrees m and n . Multiplying by powers of x repairs the degrees, and the identity then holds in general.

Theorem 8.10.4 — AB and BA Have the Same Non-zero Eigenvalues

Let $m, n \geq 1$, $A \in M_{m \times n}(F)$ and $B \in M_{n \times m}(F)$. Then

$$x^n p_{AB}(x) = x^m p_{BA}(x).$$

Consequently:

- (a) for every $\lambda \neq 0$ in F , $a_{AB}(\lambda) = a_{BA}(\lambda)$; in particular AB and BA have the same non-zero eigenvalues;
- (b) if $m = n$, then $p_{AB} = p_{BA}$; and if $m \geq n$, then $p_{AB} = x^{m-n} p_{BA}$.

Idea. We want one big matrix that “contains” AB and another that “contains” BA , and that are similar to each other. Put them in $(m+n) \times (m+n)$ block form:

$$M = \begin{pmatrix} AB & 0 \\ B & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 0 & 0 \\ B & BA \end{pmatrix}.$$

Both are block triangular, so their characteristic polynomials are $p_{AB} \cdot x^n$ and $x^m \cdot p_{BA}$. The invertible block matrix $S = \begin{pmatrix} I_m & A \\ 0 & I_n \end{pmatrix}$ turns one into the other: MS and SN are both $\begin{pmatrix} AB & ABA \\ B & BA \end{pmatrix}$.

Proof. Let $M, N, S \in M_{m+n}(F)$ be the block matrices displayed in the Idea, partitioned by m, n in rows and columns. By Theorem 7.1.1,

$$MS = \begin{pmatrix} AB \cdot I_m + 0 & AB \cdot A + 0 \\ B \cdot I_m + 0 & B \cdot A + 0 \end{pmatrix} = \begin{pmatrix} AB & ABA \\ B & BA \end{pmatrix}, \quad SN = \begin{pmatrix} 0 + AB & 0 + A \cdot BA \\ 0 + B & 0 + BA \end{pmatrix} = \begin{pmatrix} AB & ABA \\ B & BA \end{pmatrix}$$

using associativity (Theorem 0.9.4). So $MS = SN$. The matrix S is invertible: by Theorem 7.1.1, $\begin{pmatrix} I_m & A \\ 0 & I_n \end{pmatrix} \begin{pmatrix} I_m & -A \\ 0 & I_n \end{pmatrix} = \begin{pmatrix} I_m & -A + A \\ 0 & I_n \end{pmatrix} = I_{m+n}$, and Theorem 2.4.5 applies. Hence $S^{-1}MS = N$, and $p_M = p_N$ by Theorem 6.10.3.

Now compute both sides. The matrix $xI_{m+n} - M = \begin{pmatrix} xI_m - AB & 0 \\ -B & xI_n \end{pmatrix}$ is lower block triangular with square diagonal blocks, so by Theorem 6.8.4, which holds over the commutative ring $F[x]$ (the remarks before Example (A 3×3 Characteristic Polynomial)), and by Theorem 6.4.6 for xI_n ,

$$p_M(x) = \det(xI_m - AB) \det(xI_n) = p_{AB}(x) x^n.$$

Likewise $xI_{m+n} - N = \begin{pmatrix} xI_m & 0 \\ -B & xI_n - BA \end{pmatrix}$ gives $p_N(x) = x^m p_{BA}(x)$. Since $p_M = p_N$, this proves $x^n p_{AB} = x^m p_{BA}$.

- (a) Let $\lambda \neq 0$. Then λ is not a root of x^n , so $\text{mult}_\lambda(x^n) = 0$, and similarly for x^m . By Theorem 5.4.2,

$$a_{AB}(\lambda) = \text{mult}_\lambda(x^n p_{AB}) = \text{mult}_\lambda(x^m p_{BA}) = a_{BA}(\lambda).$$

A scalar is an eigenvalue exactly when its algebraic multiplicity is positive (Theorem 8.2.1, Definition 5.4.1), so the non-zero eigenvalues agree.

- (b) If $m \geq n$, then $x^n p_{AB} = x^n (x^{m-n} p_{BA})$, and canceling the non-zero polynomial x^n (Corollary 0.8.4) gives $p_{AB} = x^{m-n} p_{BA}$. For $m = n$ this is $p_{AB} = p_{BA}$.

■

This pays off the promise of Exercise (B3: AB and BA): $p_{AB} = p_{BA}$ for **all** square A and B , invertible or not. Comparing coefficients (Theorem 6.10.2) recovers $\text{tr}(AB) = \text{tr}(BA)$ and $\det(AB) = \det(BA)$ once more.

The eigenvectors travel between the two products as well, and by hand. If $AB\mathbf{w} = \lambda\mathbf{w}$ with $\lambda \neq 0$, then

$$BA(B\mathbf{w}) = B(AB\mathbf{w}) = \lambda B\mathbf{w},$$

and $B\mathbf{w} \neq \mathbf{0}$, since $A(B\mathbf{w}) = \lambda\mathbf{w} \neq \mathbf{0}$. So $B\mathbf{w}$ is an eigenvector of BA for the same λ : multiplying an eigenvector of AB by B produces one of BA . For $\lambda = 0$ the recipe can fail, because $B\mathbf{w}$ may be $\mathbf{0}$ — which is the warning below in another guise.

A rank-one example. Let $\mathbf{u}, \mathbf{v} \in F^n$ with $n \geq 2$. Take $A = \mathbf{u} \in M_{n \times 1}(F)$ and $B = \mathbf{v}^\top \in M_{1 \times n}(F)$. Then $AB = \mathbf{u}\mathbf{v}^\top$ is $n \times n$, and $BA = \mathbf{v}^\top\mathbf{u}$ is the 1×1 matrix whose entry is the number $c = \mathbf{v}^\top\mathbf{u}$, so $p_{BA} = x - c$. Applying the theorem with its m equal to our n , and its n equal to 1, gives $x p_{\mathbf{u}\mathbf{v}^\top} = x^n(x - c)$, and canceling x ,

$$p_{\mathbf{u}\mathbf{v}^\top}(x) = x^{n-1}(x - \mathbf{v}^\top\mathbf{u}).$$

For instance, the all-ones matrix $\mathbf{1}\mathbf{1}^\top \in M_n(\mathbb{R})$, with $\mathbf{1} = (1, \dots, 1)$, has $p = x^{n-1}(x - n)$: eigenvalue 0 with multiplicity $n - 1$, and eigenvalue n . Evaluating at -1 recovers Theorem 6.8.5: $\det(I + \mathbf{u}\mathbf{v}^\top) = (-1)^n \det(-I - \mathbf{u}\mathbf{v}^\top) = (-1)^n p_{\mathbf{u}\mathbf{v}^\top}(-1) = (-1)^n (-1)^{n-1} (-1 - \mathbf{v}^\top\mathbf{u}) = 1 + \mathbf{v}^\top\mathbf{u}$.

Warning. The zero eigenvalue behaves differently, and AB need not be similar to BA . Take $A = E_{12}$ and $B = E_{11}$ in $M_2(F)$. Then $AB = E_{12}E_{11} = 0$ and $BA = E_{11}E_{12} = E_{12}$. Both have $p = x^2$, as the theorem demands. But $g_{AB}(0) = 2$ while $g_{BA}(0) = 1$, and $AB = 0$ is similar only to 0, so $AB \not\sim BA$. For rectangular matrices even the algebraic multiplicities of 0 differ, by $|m - n|$.

Quick check. Let $A \in M_{2 \times 5}(\mathbb{R})$ and $B \in M_{5 \times 2}(\mathbb{R})$ with $p_{AB} = x^2 - 3x + 2$. What is p_{BA} , and what are $\text{tr}(BA)$ and $\det(BA)$?

Solution. By Theorem 8.10.4 with $m = 2$, $n = 5$, $x^5 p_{AB} = x^2 p_{BA}$, so $p_{BA} = x^3(x^2 - 3x + 2) = x^5 - 3x^4 + 2x^3$. By Theorem 6.10.2, $\text{tr}(BA) = 3$ (equal to $\text{tr}(AB)$), and $\det(BA) = (-1)^5 \cdot 0 = 0$, as it must be, since BA has rank at most $2 < 5$.

8.10.4 Left eigenvectors

An eigenvector multiplies A from the right: $A\mathbf{v} = \lambda\mathbf{v}$. Row vectors can multiply from the left, and in applications, especially Markov chains, the row version is often the one with a meaning.

A left eigenvector is a row that A only stretches when it acts from the right of the row.

Definition 8.10.1 — Left Eigenvector

Let $A \in M_n(F)$ and $\lambda \in F$. A **left eigenvector** of A for λ is a **non-zero** $\mathbf{y} \in F^n$ with

$$\mathbf{y}^\top A = \lambda \mathbf{y}^\top.$$

To distinguish them, the eigenvectors of Definition 8.2.1 are also called **right eigenvectors**.

In words: $\mathbf{y}^\top$ is a $1 \times n$ row, $\mathbf{y}^\top A$ is again a row, and we ask that it be a multiple of $\mathbf{y}^\top$. For example, $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$ and $\mathbf{y} = (1, 1)$ give $\mathbf{y}^\top A = (4 \ 4) = 4\mathbf{y}^\top$, so $(1, 1)$ is a left eigenvector for 4, whereas the right eigenvectors for 4 are the multiples of $(2, 3)$ (Example (Eigenvalues and Eigenspaces of a 2×2 Matrix)). For a diagonal matrix, $\mathbf{e}_i$ is both a left and a right eigenvector for the i -th diagonal entry. And $\mathbf{y} = \mathbf{0}$ always satisfies the equation, which is why it is excluded, exactly as for right eigenvectors.

Non-example by minimal change. For the same A , change $\mathbf{y} = (1, 1)$ to $\mathbf{y} = (1, -1)$. Then $\mathbf{y}^\top A = (1 - 3 \ 2 - 2) = (-2 \ 0)$, which is not a multiple of $(1 \ -1)$: the row is non-zero, but the defining equation $\mathbf{y}^\top A = \lambda \mathbf{y}^\top$ has no solution λ . Note that $(1, -1)$ is a right eigenvector of A , for -1 ; the two notions do not agree.

Transposing turns the definition into one we know.

Proposition 8.10.5 — Left Eigenvectors Are Eigenvectors of the Transpose

Let $A \in M_n(F)$ and $\lambda \in F$.

- (a) $\mathbf{y}$ is a left eigenvector of A for λ if and only if $\mathbf{y}$ is an eigenvector of $A^\top$ for λ .
- (b) A has a left eigenvector for λ if and only if $\lambda \in \text{spec}(A)$. Moreover $p_{A^\top} = p_A$, and $a_{A^\top}(\lambda) = a_A(\lambda)$, $g_{A^\top}(\lambda) = g_A(\lambda)$.

Proof.

- (a) By Theorem 0.9.5, $(\mathbf{y}^\top A)^\top = A^\top \mathbf{y}$ and $(\lambda \mathbf{y}^\top)^\top = \lambda \mathbf{y}$. Since transposing twice gives back the original, $\mathbf{y}^\top A = \lambda \mathbf{y}^\top$ if and only if $A^\top \mathbf{y} = \lambda \mathbf{y}$.
- (b) By Exercise (B2: Transpose), $p_{A^\top} = p_A$, so $a_{A^\top}(\lambda) = a_A(\lambda)$, and $\text{spec}(A^\top) = \text{spec}(A)$ by Theorem 8.2.1. With (a), A has a left eigenvector for λ if and only if $\lambda \in \text{spec}(A^\top) = \text{spec}(A)$. Finally, $A^\top - \lambda I = (A - \lambda I)^\top$, which has the same rank as $A - \lambda I$ (Theorem 2.5.5), so by Theorem 2.5.7, $g_{A^\top}(\lambda) = n - \text{rank}(A - \lambda I)^\top = n - \text{rank}(A - \lambda I) = g_A(\lambda)$.

■

Remark. For an operator T on V , left eigenvectors correspond to eigenvectors of the dual map T' on V^* (Chapter 4), whose matrix in the dual bases is the transpose (Theorem 4.4.2): an eigenvector of T' is a functional $\varphi \neq 0$ with $\varphi \circ T = \lambda \varphi$.

Left and right eigenvectors for the **same** eigenvalue are usually unrelated, as $(1, 1)$ and $(2, 3)$ show. For **different** eigenvalues, though, they are always orthogonal in the sense of the plain product $\mathbf{y}^\top \mathbf{v}$.

Theorem 8.10.6 — Left and Right Eigenvectors Are Biorthogonal

Let $A \in M_n(F)$, let $\mathbf{y}$ be a left eigenvector of A for μ , and let $\mathbf{v}$ be a right eigenvector of A for λ . If $\lambda \neq \mu$, then $\mathbf{y}^\top \mathbf{v} = 0$.

Proof. Compute the 1×1 matrix $\mathbf{y}^T A \mathbf{v}$ in two ways, using associativity (Theorem 0.9.4):

$$\mu \mathbf{y}^T \mathbf{v} = (\mathbf{y}^T A) \mathbf{v} = \mathbf{y}^T A \mathbf{v} = \mathbf{y}^T (A \mathbf{v}) = \lambda \mathbf{y}^T \mathbf{v}.$$

Hence $(\lambda - \mu) \mathbf{y}^T \mathbf{v} = 0$. Since $\lambda - \mu \neq 0$, this forces $\mathbf{y}^T \mathbf{v} = 0$ (Theorem 0.7.1). ■

For the example, $(1, 1)$ is a left eigenvector for 4 and $(1, -1)$ a right eigenvector for -1 , and indeed $1 - 1 = 0$.

For a diagonalizable matrix the left eigenvectors come for free with the diagonalization, and they give a formula for all powers that we use in Section 11.

Proposition 8.10.7 — Expansion in Left and Right Eigenvectors

Let $A = PDP^{-1}$ with $P \in M_n(F)$ invertible and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be the columns of P , and let $\mathbf{y}_1^T, \dots, \mathbf{y}_n^T$ be the rows of P^{-1} . Then:

- (a) $\mathbf{v}_i$ is a right eigenvector and $\mathbf{y}_i$ a left eigenvector of A for λ_i ;
- (b) $\mathbf{y}_i^T \mathbf{v}_j = 1$ if $i = j$ and 0 if $i \neq j$;
- (c) for every $q \in F[x]$,

$$q(A) = \sum_{i=1}^n q(\lambda_i) \mathbf{v}_i \mathbf{y}_i^T, \quad \text{in particular} \quad A^k = \sum_{i=1}^n \lambda_i^k \mathbf{v}_i \mathbf{y}_i^T \quad (k \in \mathbb{N}).$$

Proof.

- (a) From $AP = PD$, column i reads $A \mathbf{v}_i = \lambda_i \mathbf{v}_i$ (Section 4), and $\mathbf{v}_i \neq \mathbf{0}$ because P is invertible. From $P^{-1}A = DP^{-1}$, row i reads $\mathbf{y}_i^T A = \lambda_i \mathbf{y}_i^T$ (Theorem 0.9.2), and $\mathbf{y}_i \neq \mathbf{0}$ because P^{-1} is invertible.
- (b) The (i, j) -entry of $P^{-1}P = I$ is row i of P^{-1} times column j of P , that is, $\mathbf{y}_i^T \mathbf{v}_j$.
- (c) By Theorem 8.4.3, $q(A) = P q(D) P^{-1}$. The columns of $P q(D)$ are $q(\lambda_i) \mathbf{v}_i$, and the rows of P^{-1} are $\mathbf{y}_i^T$, so Corollary 7.1.3 gives $q(A) = \sum_i q(\lambda_i) \mathbf{v}_i \mathbf{y}_i^T$. Take $q = x^k$. ■

Part (b) is biorthogonality with a normalization: $\mathbf{y}_i^T \mathbf{v}_i = 1$. Part (c) splits A into rank-one pieces $\mathbf{v}_i \mathbf{y}_i^T$, one per eigenvalue, and each piece is simply multiplied by λ_i^k when A is raised to the power k . In the running example, $P = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}$ and $D = \text{diag}(4, -1)$ give $P^{-1} = \frac{1}{5} \begin{pmatrix} 1 & 1 \\ 3 & -2 \end{pmatrix}$, so $\mathbf{y}_1 = \frac{1}{5}(1, 1)$ and $\mathbf{y}_2 = \frac{1}{5}(3, -2)$, and

$$A^k = \frac{4^k}{5} \begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix} + \frac{(-1)^k}{5} \begin{pmatrix} 3 & -2 \\ -3 & 2 \end{pmatrix}.$$

For $k = 1$ this is $\frac{1}{5} \begin{pmatrix} 8-3 & 8+2 \\ 12+3 & 12-2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$, as it should be.

8.10.5 Exercises

8.10.5.1 A. Check your understanding

Exercise A1.

- State the Spectral Mapping Theorem, including its hypothesis on p_T .
- True or false: for every $A \in M_n(\mathbb{R})$, every real eigenvalue of A^2 is the square of a real eigenvalue of A . Justify your answer.
- True or false: if $A \in M_{2 \times 3}(F)$ and $B \in M_{3 \times 2}(F)$, then $p_{AB} = p_{BA}$. Justify your answer.
- True or false: for all $A, B \in M_n(F)$, AB is similar to BA . Justify your answer.
- Define a left eigenvector of $A \in M_n(F)$, and say how left eigenvectors of A are related to A^T .
- True or false: a matrix $A \in M_3(\mathbb{C})$ with $\operatorname{tr} A = \operatorname{tr} A^2 = \operatorname{tr} A^3 = 0$ satisfies $A^3 = 0$. Justify your answer.

Solution.

- If V is finite-dimensional with $\dim V = n \geq 1$, $T \in \mathcal{L}(V)$, and $p_T = \prod_{i=1}^n (x - \lambda_i)$ splits over F , then for every $q \in F[x]$, $p_{q(T)} = \prod_i (x - q(\lambda_i))$ and $\operatorname{spec}(q(T)) = q(\operatorname{spec}(T))$ ([Theorem 8.10.1](#)).
- False. The rotation $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has no real eigenvalue, while $R^2 = -I$ has the real eigenvalue -1 .
- False. They have different degrees, 2 and 3. The correct identity is $x^3 p_{AB} = x^2 p_{BA}$, that is, $p_{BA} = x p_{AB}$ ([Theorem 8.10.4](#)).
- False. $A = E_{12}$, $B = E_{11}$ give $AB = 0$ and $BA = E_{12} \neq 0$, and the only matrix similar to 0 is 0.
- A non-zero $\mathbf{y} \in F^n$ with $\mathbf{y}^T A = \lambda \mathbf{y}^T$ for some $\lambda \in F$. Equivalently, $\mathbf{y}$ is an eigenvector of A^T for λ ([Proposition 8.10.5](#)).
- True. By [Theorem 8.10.3](#) with $n = 3$, (e) $\Rightarrow$ (d).

8.10.5.2 B. Practice

Exercise B1: Eigenvalues of a polynomial in A .

Let $A \in M_3(\mathbb{C})$ have $p_A = (x - 1)^2(x + 2)$.

- Find p_{A^2+A+I} , $\operatorname{tr}(A^2 + A + I)$ and $\det(A^2 + A + I)$. Is $A^2 + A + I$ invertible?
- Find p_{A^2+A-2I} . Hence show that $A^2 + A - 2I$ is nilpotent.
- Find $\operatorname{tr} A^5$ and $\det(A^{-1})$.

Solution. The list of eigenvalues of A with multiplicity is $1, 1, -2$, since p_A splits.

- With $q = x^2 + x + 1$, $q(1) = 3$ and $q(-2) = 4 - 2 + 1 = 3$. By [Theorem 8.10.1](#), $p_{q(A)} = (x - 3)^3$, and by [Corollary 8.10.2](#), $\operatorname{tr} q(A) = 9$ and $\det q(A) = 27 \neq 0$. So

$A^2 + A + I$ is invertible (Theorem 6.5.3).

- (b) With $r = x^2 + x - 2 = (x - 1)(x + 2)$, $r(1) = r(-2) = 0$, so $p_{r(A)} = x^3$ by Theorem 8.10.1. By Theorem 8.10.3 ((c) $\Rightarrow$ (d)), $r(A)^3 = 0$, so $A^2 + A - 2I$ is nilpotent.
- (c) By Corollary 8.10.2, $\text{tr } A^5 = 1 + 1 + (-2)^5 = -30$. Also $\det A = 1 \cdot 1 \cdot (-2) = -2 \neq 0$ (Theorem 8.3.3), so A is invertible and $\det(A^{-1}) = (\det A)^{-1} = -\frac{1}{2}$ (Corollary 6.5.6).

Exercise B2: AB and BA for a 2×3 example.

Let $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \\ 1 & 1 \end{pmatrix}$.

- (a) Compute AB , BA , p_{AB} and p_{BA} , and verify that $x^3 p_{AB} = x^2 p_{BA}$.
- (b) Hence list the eigenvalues of BA without further computation, and find an eigenvector of BA for each non-zero eigenvalue by applying B to eigenvectors of AB .

Solution.

- (a) By row-times-column,

$$AB = \begin{pmatrix} 2 & 1 \\ 0 & -2 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 2 & 1 \end{pmatrix}.$$

AB is upper triangular, so $p_{AB} = (x - 2)(x + 2) = x^2 - 4$. Expanding $\det(xI - BA)$ along the second row, whose only non-zero entry is $x + 2$ in position $(2, 2)$,

$$p_{BA} = (x + 2) \det \begin{pmatrix} x - 1 & -1 \\ -1 & x - 1 \end{pmatrix} = (x + 2)((x - 1)^2 - 1) = (x + 2)x(x - 2) = x^3 - 4x.$$

Then $x^3 p_{AB} = x^5 - 4x^3 = x^2(x^3 - 4x) = x^2 p_{BA}$.

- (b) The eigenvalues of BA are the roots of $x(x - 2)(x + 2)$: $0, 2, -2$, matching Theorem 8.10.4 (the non-zero ones are those of AB). If $AB\mathbf{w} = \lambda\mathbf{w}$, then $BA(B\mathbf{w}) = B(AB\mathbf{w}) = \lambda B\mathbf{w}$, so $B\mathbf{w}$ is an eigenvector of BA as long as $B\mathbf{w} \neq \mathbf{0}$. For $\lambda = 2$: $\mathbf{w} = (1, 0)$, $B\mathbf{w} = (1, 0, 1)$, and $BA(1, 0, 1) = (2, 0, 2)$. For $\lambda = -2$: $AB + 2I = \begin{pmatrix} 4 & 1 \\ 0 & 0 \end{pmatrix}$ gives $\mathbf{w} = (1, -4)$, $B\mathbf{w} = (1, 4, -3)$, and $BA(1, 4, -3) = (1 - 3, -8, 1 + 8 - 3) = (-2, -8, 6) = -2(1, 4, -3)$.

Exercise B3: Left eigenvectors of a 2×2 matrix.

Let $A = \begin{pmatrix} 0 & 1 \\ -6 & 5 \end{pmatrix} \in M_2(\mathbb{R})$, whose right eigenvectors $(1, 2)$ for 2 and $(1, 3)$ for 3 were found in Exercise (B1: Diagonalize or show impossible).

- (a) Find a left eigenvector of A for each eigenvalue.
- (b) Verify Theorem 8.10.6 for these vectors, and rescale the left eigenvectors so that

$$\mathbf{y}_i^\top \mathbf{v}_i = 1.$$

- (c) Hence write A^k as a combination of two rank-one matrices, and check the formula for $k = 0$ and $k = 1$.

Solution.

- (a) By [Proposition 8.10.5](#), we need eigenvectors of $A^\top = \begin{pmatrix} 0 & -6 \\ 1 & 5 \end{pmatrix}$. For 2: $A^\top - 2I = \begin{pmatrix} -2 & -6 \\ 1 & 3 \end{pmatrix}$ gives $y_1 = -3y_2$, so $(3, -1)$. For 3: $A^\top - 3I = \begin{pmatrix} -3 & -6 \\ 1 & 2 \end{pmatrix}$ gives $y_1 = -2y_2$, so $(2, -1)$. Check: $\begin{pmatrix} 3 & -1 \end{pmatrix} A = \begin{pmatrix} 6 & -2 \end{pmatrix}$ and $\begin{pmatrix} 2 & -1 \end{pmatrix} A = \begin{pmatrix} 6 & -3 \end{pmatrix}$.
- (b) $\begin{pmatrix} 3 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 3 \end{pmatrix} = 0$ and $\begin{pmatrix} 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \end{pmatrix} = 0$, as the theorem predicts for different eigenvalues. Also $\begin{pmatrix} 3 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \end{pmatrix} = 1$ and $\begin{pmatrix} 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 3 \end{pmatrix} = -1$, so take $\mathbf{y}_1 = \begin{pmatrix} 3 & -1 \end{pmatrix}$ and $\mathbf{y}_2 = \begin{pmatrix} -2 & 1 \end{pmatrix}$.
- (c) With $P = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$, the rows of $P^{-1} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix}$ are exactly $\mathbf{y}_1^\top, \mathbf{y}_2^\top$, and by [Proposition 8.10.7](#),

$$A^k = 2^k \begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 3 & -1 \end{pmatrix} + 3^k \begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} -2 & 1 \end{pmatrix} = 2^k \begin{pmatrix} 3 & -1 \\ 6 & -2 \end{pmatrix} + 3^k \begin{pmatrix} -2 & 1 \\ -6 & 3 \end{pmatrix}.$$

$$\text{For } k = 0: \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \text{ For } k = 1: \begin{pmatrix} 6 - 6 & -2 + 3 \\ 12 - 18 & -4 + 9 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -6 & 5 \end{pmatrix} = A.$$

8.10.5.3 C. Going deeper

Exercise C1: Traces of powers of AB and BA .

Let $A \in M_{m \times n}(F)$ and $B \in M_{n \times m}(F)$.

- (a) Prove that $B(AB)^{k-1}A = (BA)^k$ for every $k \geq 1$, and deduce that $\text{tr}((AB)^k) = \text{tr}((BA)^k)$.
- (b) Now let $F = \mathbb{C}$. Give a second proof of the trace identity from [Theorem 8.10.4](#) and [Corollary 8.10.2](#).

Solution.

- (a) By induction on k . For $k = 1$ both sides are BA . If $B(AB)^{k-1}A = (BA)^k$, then by associativity

$$B(AB)^k A = B(AB)^{k-1}(AB)A = (B(AB)^{k-1}A)(BA) = (BA)^k(BA) = (BA)^{k+1}.$$

Now $(AB)^k = A(B(AB)^{k-1})$, with A of size $m \times n$ and $B(AB)^{k-1}$ of size $n \times m$. By [Theorem 0.9.6 \(c\)](#),

$$\text{tr}((AB)^k) = \text{tr}(A \cdot B(AB)^{k-1}) = \text{tr}(B(AB)^{k-1} \cdot A) = \text{tr}((BA)^k).$$

- (b) Over $\mathbb{C}$, p_{AB} and p_{BA} split. Suppose $m \geq n$ (otherwise swap the roles of A and B). By [Theorem 8.10.4](#), $p_{AB} = x^{m-n}p_{BA}$, so if $\lambda_1, \dots, \lambda_n$ is the list of eigenvalues of BA with multiplicity, then the list for AB is $\lambda_1, \dots, \lambda_n$ together with $m - n$

zeros. For $k \geq 1$ the zeros contribute $0^k = 0$, so by [Corollary 8.10.2](#), $\operatorname{tr}((AB)^k) = \sum_i \lambda_i^k = \operatorname{tr}((BA)^k)$.

Exercise C2: Adding a commuting nilpotent matrix.

Let $A, N \in M_n(\mathbb{C})$, $n \geq 1$, with $AN = NA$ and N nilpotent.

- (a) Prove that $p_{A+N} = p_A$; in particular $\operatorname{spec}(A + N) = \operatorname{spec}(A)$ and $\det(A + N) = \det A$.
- (b) Deduce that if A is invertible, so is $A + N$.
- (c) Show that (a) fails without $AN = NA$.

Hint: triangularize A and N together.

Solution.

- (a) By Theorem 8.9.4 there is an invertible P with $U = P^{-1}AP$ and $U' = P^{-1}NP$ upper triangular. Since N is nilpotent, $\operatorname{spec}(N) = \{0\}$ (Exercise (C2: Nilpotent operators)), so every diagonal entry of U' is 0 by Theorem 8.7.4. Then $P^{-1}(A + N)P = U + U'$ is upper triangular with the same diagonal as U . By Theorem 6.10.3 and the examples after Definition 6.10.1,

$$p_{A+N} = p_{U+U'} = \prod_i (x - u_{ii}) = p_U = p_A.$$

Equal characteristic polynomials have the same roots (Theorem 8.2.1) and the same constant term $(-1)^n \det$ (Theorem 6.10.2).

- (b) By (a), $\det(A + N) = \det A \neq 0$, so $A + N$ is invertible (Theorem 6.5.3).
- (c) Let $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Both are nilpotent, so $p_A = x^2$, but $A + N = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has $p = x^2 - 1$ and eigenvalues ± 1 . Here $AN = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = NA$.

Exercise C3: How many traces, and which field?.

- (a) Let $\omega = e^{2\pi i/3} \in \mathbb{C}$ and $C = \operatorname{diag}(1, \omega, \omega^2)$. Show that $\operatorname{tr} C = \operatorname{tr} C^2 = 0$ but C is not nilpotent. Explain why this does not contradict [Theorem 8.10.3](#).
- (b) Let $n \geq 2$. Give a matrix $C \in M_n(\mathbb{C})$ with $\operatorname{tr} C^k = 0$ for $k = 1, \dots, n-1$ that is not nilpotent.
- (c) Let $A, B \in M_n(\mathbb{C})$ and suppose $C = AB - BA$ commutes with A . Prove that $C^n = 0$.
- (d) Over $\mathbb{F}_2$, find a non-nilpotent $C \in M_2(\mathbb{F}_2)$ with $\operatorname{tr} C^k = 0$ for every $k \geq 1$.

Solution.

- (a) $\omega^3 = 1$ and $\omega \neq 1$, so $0 = \omega^3 - 1 = (\omega - 1)(\omega^2 + \omega + 1)$ gives $1 + \omega + \omega^2 = 0$. Hence $\operatorname{tr} C = 1 + \omega + \omega^2 = 0$, and $\operatorname{tr} C^2 = 1 + \omega^2 + \omega^4 = 1 + \omega^2 + \omega = 0$. But $C^k = \operatorname{diag}(1, \omega^k, \omega^{2k}) \neq 0$ for every k . No contradiction: the theorem, for $n = 3$,

requires $\operatorname{tr} C^k = 0$ for $k = 1, 2, 3$, and $\operatorname{tr} C^3 = 1 + 1 + 1 = 3 \neq 0$.

- (b) Let $\zeta = e^{2\pi i/n}$ and $C = \operatorname{diag}(1, \zeta, \dots, \zeta^{n-1})$. For $1 \leq k \leq n-1$, $\zeta^k \neq 1$ and $(\zeta^k)^n = 1$, so $\operatorname{tr} C^k = \sum_{j=0}^{n-1} (\zeta^k)^j = \frac{(\zeta^k)^n - 1}{\zeta^k - 1} = 0$ by the geometric sum. But C^k has the entry 1 in position $(1, 1)$ for all k , so C is not nilpotent.
- (c) By Exercise (C2: When A commutes with $[A, B]$) (a), $\operatorname{tr} C^k = 0$ for every $k \geq 1$, in particular for $k = 1, \dots, n$. By [Theorem 8.10.3](#) ((e) $\Rightarrow$ (d)), $C^n = 0$.
- (d) $C = I_2$: $\operatorname{tr} C^k = 1 + 1 = 0$ in $\mathbb{F}_2$ for every k , and $C^k = I_2 \neq 0$.